Sales Team and Inventory Analysis

SQL project | SQLite | Sales of 2025

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This project uses a database of a home-appliance company (sales reps, products and sales) to measure rep performance against targets, find best sellers and check stock levels. It practises LEFT JOIN, CASE, RANK, ROW_NUMBER, CTEs and subqueries. All money values are in Rs. The data is sample data created for this project.

Database tables

Table	Columns	Rows
employees	emp_id, name, region, monthly_target	12
products	product_id, name, category, price, stock, reorder_level	14
sales	sale_id, emp_id, product_id, sale_date, quantity, amount	520

Key findings

• 5 reps exceeded their annual target, 4 were near target (90%+) and 3 were below. Best: Ayesha Butt (122.8%); lowest: Hamza Sheikh (88.2%).

• Top rep in each region: East: Usman Khan (Rs 2,996,785); North: Imran Chaudhry (Rs 3,170,335); South: Noor Ansari (Rs 2,522,275); West: Kiran Raza (Rs 2,540,870).

• Q3 was the strongest quarter (Rs 9,563,715) and Q1 the weakest (Rs 5,379,080). Total for the year: Rs 26,707,135.

• Best seller by revenue: Large Appliances: Air Conditioner; Kitchen: Microwave Oven; Home Care: Vacuum Cleaner.

• 6 products are at or below their reorder level. The most urgent is Electric Kettle (stock 11, 45 units sold in the last 90 days).

• Thursday has the highest sales (Rs 5,245,885) and Wednesday the lowest (Rs 2,800,360).

• 1 product(s) had no sales at all: Steam Cleaner. Review pricing, promotion or whether it should still be stocked.

Analysis queries

Q1. Sales rep performance against annual target

LEFT JOIN, aggregate functions and a CASE expression to label each rep.

```
SELECT e.name, e.region,
       COUNT(s.sale_id)                    AS deals,
       ROUND(SUM(s.amount), 0)             AS total_sales,
       e.monthly_target * 12               AS annual_target,
       ROUND(100.0 * SUM(s.amount) / (e.monthly_target * 12), 1) AS achievement_pct,
       CASE WHEN SUM(s.amount) >= e.monthly_target * 12        THEN 'Exceeded'
            WHEN SUM(s.amount) >= e.monthly_target * 12 * 0.9  THEN 'Near target'
            ELSE 'Below target' END        AS status
FROM employees e
LEFT JOIN sales s ON s.emp_id = e.emp_id
GROUP BY e.emp_id
ORDER BY achievement_pct DESC;
```

Name	Region	Deals	Total sales	Annual target	Achievement pct	Status
Ayesha Butt	South	57	2,505,975	2,040,000	122.8	Exceeded
Sana Malik	South	42	1,799,770	1,560,000	115.4	Exceeded
Saad Hussain	East	28	1,631,185	1,440,000	113.3	Exceeded
Ali Raza	North	43	2,409,740	2,280,000	105.7	Exceeded
Imran Chaudhry	North	55	3,170,335	3,120,000	101.6	Exceeded
Maryam Qureshi	West	23	1,030,490	1,080,000	95.4	Near target
Kiran Raza	West	51	2,540,870	2,760,000	92.1	Near target
Noor Ansari	South	49	2,522,275	2,760,000	91.4	Near target
Bilal Ahmed	North	56	2,378,245	2,640,000	90.1	Near target
Usman Khan	East	44	2,996,785	3,360,000	89.2	Below target
Hira Siddiqui	West	31	1,497,920	1,680,000	89.2	Below target
Hamza Sheikh	East	41	2,223,545	2,520,000	88.2	Below target

Insight: 5 reps exceeded their annual target, 4 were near target (90%+) and 3 were below. Best: Ayesha Butt (122.8%); lowest: Hamza Sheikh (88.2%).

Q2. Rank of each rep inside their region

RANK() with PARTITION BY restarts the ranking for every region.

```
SELECT region, name, total_sales,
       RANK() OVER (PARTITION BY region ORDER BY total_sales DESC) AS rank_in_region
FROM (
    SELECT e.region, e.name, ROUND(SUM(s.amount), 0) AS total_sales
    FROM employees e
    JOIN sales s ON s.emp_id = e.emp_id
    GROUP BY e.emp_id
)
ORDER BY region, rank_in_region;
```

Region	Name	Total sales	Rank in region
East	Usman Khan	2,996,785	1
East	Hamza Sheikh	2,223,545	2
East	Saad Hussain	1,631,185	3
North	Imran Chaudhry	3,170,335	1
North	Ali Raza	2,409,740	2
North	Bilal Ahmed	2,378,245	3
South	Noor Ansari	2,522,275	1
South	Ayesha Butt	2,505,975	2
South	Sana Malik	1,799,770	3
West	Kiran Raza	2,540,870	1
West	Hira Siddiqui	1,497,920	2
West	Maryam Qureshi	1,030,490	3

Insight: Top rep in each region: East: Usman Khan (Rs 2,996,785); North: Imran Chaudhry (Rs 3,170,335); South: Noor Ansari (Rs 2,522,275); West: Kiran Raza (Rs 2,540,870).

Q3. Quarterly sales with cumulative total

Groups dates into quarters with strftime and adds a running total.

```
WITH q AS (
    SELECT 'Q' || ((CAST(strftime('%m', sale_date) AS INTEGER) + 2) / 3) AS quarter,
           SUM(amount) AS sales
    FROM sales
    GROUP BY quarter
)
SELECT quarter,
       ROUND(sales, 0) AS sales,
       ROUND(SUM(sales) OVER (ORDER BY quarter), 0) AS cumulative
FROM q
ORDER BY quarter;
```

Quarter	Sales	Cumulative
Q1	5,379,080	5,379,080
Q2	6,173,565	11,552,645
Q3	9,563,715	21,116,360
Q4	5,590,775	26,707,135

Insight: Q3 was the strongest quarter (Rs 9,563,715) and Q1 the weakest (Rs 5,379,080). Total for the year: Rs 26,707,135.

Q4. Best-selling product in each category

ROW_NUMBER() with PARTITION BY picks the number one product per category.

```
WITH product_sales AS (
    SELECT p.category, p.name AS product,
           SUM(s.quantity) AS units,
           ROUND(SUM(s.amount), 0) AS revenue
    FROM sales s
    JOIN products p ON p.product_id = s.product_id
    GROUP BY p.product_id
),
ranked AS (
    SELECT *, ROW_NUMBER() OVER (PARTITION BY category ORDER BY revenue DESC) AS rn
    FROM product_sales
)
SELECT category, product, units, revenue
FROM ranked
WHERE rn = 1
ORDER BY revenue DESC;
```

Category	Product	Units	Revenue
Large Appliances	Air Conditioner	51	6,669,000
Kitchen	Microwave Oven	73	1,988,000
Home Care	Vacuum Cleaner	62	1,441,200

Insight: Best seller by revenue: Large Appliances: Air Conditioner; Kitchen: Microwave Oven; Home Care: Vacuum Cleaner.

Q5. Stock alert: products at or below reorder level

Finds low-stock products and checks how fast they sold in the last 90 days of data.

```
SELECT p.name, p.category, p.stock, p.reorder_level,
       COALESCE(SUM(s.quantity), 0) AS units_sold_last_90_days
FROM products p
LEFT JOIN sales s
       ON s.product_id = p.product_id
      AND s.sale_date >= date((SELECT MAX(sale_date) FROM sales), '-90 days')
WHERE p.stock <= p.reorder_level
GROUP BY p.product_id
ORDER BY units_sold_last_90_days DESC;
```

Name	Category	Stock	Reorder level	Units sold last 90 days
Electric Kettle	Kitchen	11	20	45
Air Fryer	Kitchen	9	15	23
Water Dispenser	Large Appliances	6	8	20
Air Conditioner	Large Appliances	12	15	11
Washing Machine	Large Appliances	8	10	6
Steam Cleaner	Home Care	5	8	0

Insight: 6 products are at or below their reorder level. The most urgent is Electric Kettle (stock 11, 45 units sold in the last 90 days).

Q6. Sales by day of the week

strftime('%w') gives the weekday number; CASE turns it into a name.

```
SELECT CASE strftime('%w', sale_date)
           WHEN '0' THEN 'Sunday' WHEN '1' THEN 'Monday' WHEN '2' THEN 'Tuesday' WHEN '3' THEN 'Wednesday' WHEN '4' THE
       END AS weekday,
       COUNT(*) AS deals,
       ROUND(SUM(amount), 0) AS sales
FROM sales
GROUP BY strftime('%w', sale_date)
ORDER BY strftime('%w', sale_date);
```

Weekday	Deals	Sales
Sunday	74	3,549,880
Monday	60	3,017,335
Tuesday	76	4,021,280
Wednesday	63	2,800,360
Thursday	87	5,245,885
Friday	83	4,035,660
Saturday	77	4,036,735

Insight: Thursday has the highest sales (Rs 5,245,885) and Wednesday the lowest (Rs 2,800,360).

Q7. Products that were never sold

NOT EXISTS with a correlated subquery.

```
SELECT p.product_id, p.name, p.category, p.stock
FROM products p
WHERE NOT EXISTS (
    SELECT 1 FROM sales s WHERE s.product_id = p.product_id
);
```

Product id	Name	Category	Stock
14	Steam Cleaner	Home Care	5

Insight: 1 product(s) had no sales at all: Steam Cleaner. Review pricing, promotion or whether it should still be stocked.

Appendix A: Sales list

First 45 records out of 520 sales. Query used:

```
SELECT s.sale_id, s.sale_date, e.name AS sales_rep, e.region,
       p.name AS product, s.quantity, s.amount
FROM sales s
JOIN employees e ON e.emp_id = s.emp_id
JOIN products p  ON p.product_id = s.product_id
ORDER BY s.sale_date, s.sale_id
LIMIT 45;
```

Sale id	Sale date	Sales rep	Region	Product	Quantity	Amount
1	2025-01-01	Ayesha Butt	South	Refrigerator	2	176,400
2	2025-01-02	Ayesha Butt	South	Washing Machine	1	68,400
3	2025-01-03	Noor Ansari	South	Electric Kettle	1	4,500
4	2025-01-03	Kiran Raza	West	Ceiling Fan	1	11,875
5	2025-01-04	Hira Siddiqui	West	Refrigerator	1	98,000
6	2025-01-04	Hamza Sheikh	East	Blender	2	12,960
7	2025-01-04	Noor Ansari	South	Microwave Oven	1	25,200
8	2025-01-04	Ayesha Butt	South	Toaster	1	4,940
9	2025-01-04	Ayesha Butt	South	Iron	3	10,830
10	2025-01-05	Maryam Qureshi	West	Toaster	4	20,800
11	2025-01-06	Noor Ansari	South	Ceiling Fan	5	56,250
12	2025-01-07	Ayesha Butt	South	Refrigerator	1	98,000
13	2025-01-07	Ayesha Butt	South	Ceiling Fan	2	25,000
14	2025-01-08	Usman Khan	East	Water Dispenser	3	86,400
15	2025-01-09	Maryam Qureshi	West	Air Conditioner	2	243,000
16	2025-01-09	Kiran Raza	West	Electric Kettle	1	4,050
17	2025-01-10	Imran Chaudhry	North	Microwave Oven	1	28,000
18	2025-01-10	Usman Khan	East	Electric Kettle	5	21,375
19	2025-01-11	Noor Ansari	South	Toaster	5	26,000
20	2025-01-12	Imran Chaudhry	North	Blender	3	21,600
21	2025-01-12	Bilal Ahmed	North	Iron	4	15,200
22	2025-01-16	Saad Hussain	East	Vacuum Cleaner	3	72,000
23	2025-01-18	Maryam Qureshi	West	Iron	1	3,800
24	2025-01-20	Bilal Ahmed	North	Ceiling Fan	3	37,500
25	2025-01-21	Maryam Qureshi	West	Water Dispenser	2	60,800
26	2025-01-21	Kiran Raza	West	Washing Machine	1	72,000
27	2025-01-23	Sana Malik	South	Iron	4	14,440
28	2025-01-23	Noor Ansari	South	Electric Kettle	2	9,000
29	2025-01-23	Maryam Qureshi	West	Electric Kettle	1	4,500
30	2025-01-23	Noor Ansari	South	Ceiling Fan	3	33,750
31	2025-01-24	Kiran Raza	West	Toaster	2	9,360
32	2025-01-24	Bilal Ahmed	North	Microwave Oven	3	79,800

Sale id	Sale date	Sales rep	Region	Product	Quantity	Amount
33	2025-01-27	Imran Chaudhry	North	Blender	4	28,800
34	2025-01-28	Sana Malik	South	LED TV	1	61,750
35	2025-01-30	Maryam Qureshi	West	Ceiling Fan	1	12,500
36	2025-01-30	Noor Ansari	South	Refrigerator	1	98,000
37	2025-01-31	Kiran Raza	West	Toaster	3	15,600
38	2025-01-31	Sana Malik	South	Air Conditioner	2	270,000
39	2025-02-01	Ali Raza	North	Water Dispenser	2	60,800
40	2025-02-02	Hamza Sheikh	East	Ceiling Fan	4	50,000
41	2025-02-03	Bilal Ahmed	North	Iron	4	14,440
42	2025-02-03	Hira Siddiqui	West	Vacuum Cleaner	2	43,200
43	2025-02-04	Usman Khan	East	Vacuum Cleaner	1	24,000
44	2025-02-05	Kiran Raza	West	Iron	3	11,400
45	2025-02-06	Noor Ansari	South	Microwave Oven	2	53,200

Appendix B: Full SQL script

Create tables, insert data and run all queries in one file.

```
-- Sales Team and Inventory Analysis
-- Run in SQLite (sqlite3 or DB Browser for SQLite)

-- PART 1: CREATE TABLES
CREATE TABLE employees (
    emp_id         INTEGER PRIMARY KEY,
    name           TEXT,
    region         TEXT,
    monthly_target INTEGER      -- sales target per month (Rs)
);
CREATE TABLE products (
    product_id    INTEGER PRIMARY KEY,
    name          TEXT,
    category      TEXT,
    price         INTEGER,      -- Rs
    stock         INTEGER,      -- units in warehouse
    reorder_level INTEGER       -- order more when stock <= this
);
CREATE TABLE sales (
    sale_id    INTEGER PRIMARY KEY,
    emp_id     INTEGER REFERENCES employees(emp_id),
    product_id INTEGER REFERENCES products(product_id),
    sale_date  TEXT,            -- YYYY-MM-DD
    quantity   INTEGER,
    amount     REAL             -- final amount after discount (Rs)
);

-- PART 2: INSERT DATA
INSERT INTO employees VALUES (1, 'Ali Raza', 'North', 190000);
INSERT INTO employees VALUES (2, 'Sana Malik', 'South', 130000);
INSERT INTO employees VALUES (3, 'Usman Khan', 'East', 280000);
INSERT INTO employees VALUES (4, 'Hira Siddiqui', 'West', 140000);
INSERT INTO employees VALUES (5, 'Bilal Ahmed', 'North', 220000);
INSERT INTO employees VALUES (6, 'Ayesha Butt', 'South', 170000);
INSERT INTO employees VALUES (7, 'Hamza Sheikh', 'East', 210000);
INSERT INTO employees VALUES (8, 'Maryam Qureshi', 'West', 90000);
INSERT INTO employees VALUES (9, 'Imran Chaudhry', 'North', 260000);
INSERT INTO employees VALUES (10, 'Noor Ansari', 'South', 230000);
INSERT INTO employees VALUES (11, 'Saad Hussain', 'East', 120000);
INSERT INTO employees VALUES (12, 'Kiran Raza', 'West', 230000);
INSERT INTO products VALUES (1, 'Air Conditioner', 'Large Appliances', 135000, 12, 15);
INSERT INTO products VALUES (2, 'Refrigerator', 'Large Appliances', 98000, 25, 10);
INSERT INTO products VALUES (3, 'Washing Machine', 'Large Appliances', 72000, 8, 10);
INSERT INTO products VALUES (4, 'LED TV', 'Large Appliances', 65000, 30, 12);
INSERT INTO products VALUES (5, 'Water Dispenser', 'Large Appliances', 32000, 6, 8);
INSERT INTO products VALUES (6, 'Ceiling Fan', 'Large Appliances', 12500, 60, 25);
INSERT INTO products VALUES (7, 'Microwave Oven', 'Kitchen', 28000, 14, 10);
INSERT INTO products VALUES (8, 'Air Fryer', 'Kitchen', 16500, 9, 15);
INSERT INTO products VALUES (9, 'Blender', 'Kitchen', 7200, 40, 20);
INSERT INTO products VALUES (10, 'Toaster', 'Kitchen', 5200, 18, 15);
INSERT INTO products VALUES (11, 'Electric Kettle', 'Kitchen', 4500, 11, 20);
INSERT INTO products VALUES (12, 'Vacuum Cleaner', 'Home Care', 24000, 22, 10);
INSERT INTO products VALUES (13, 'Iron', 'Home Care', 3800, 50, 20);
INSERT INTO products VALUES (14, 'Steam Cleaner', 'Home Care', 19000, 5, 8);
INSERT INTO sales VALUES (1, 6, 2, '2025-01-01', 2, 176400.0);
INSERT INTO sales VALUES (2, 6, 3, '2025-01-02', 1, 68400.0);
INSERT INTO sales VALUES (3, 10, 11, '2025-01-03', 1, 4500.0);
INSERT INTO sales VALUES (4, 12, 6, '2025-01-03', 1, 11875.0);
INSERT INTO sales VALUES (5, 4, 2, '2025-01-04', 1, 98000.0);
INSERT INTO sales VALUES (6, 7, 9, '2025-01-04', 2, 12960.0);
INSERT INTO sales VALUES (7, 10, 7, '2025-01-04', 1, 25200.0);
INSERT INTO sales VALUES (8, 6, 10, '2025-01-04', 1, 4940.0);
INSERT INTO sales VALUES (9, 6, 13, '2025-01-04', 3, 10830.0);
INSERT INTO sales VALUES (10, 8, 10, '2025-01-05', 4, 20800.0);
INSERT INTO sales VALUES (11, 10, 6, '2025-01-06', 5, 56250.0);
INSERT INTO sales VALUES (12, 6, 2, '2025-01-07', 1, 98000.0);
INSERT INTO sales VALUES (13, 6, 6, '2025-01-07', 2, 25000.0);
INSERT INTO sales VALUES (14, 3, 5, '2025-01-08', 3, 86400.0);
INSERT INTO sales VALUES (15, 8, 1, '2025-01-09', 2, 243000.0);
INSERT INTO sales VALUES (16, 12, 11, '2025-01-09', 1, 4050.0);
INSERT INTO sales VALUES (17, 9, 7, '2025-01-10', 1, 28000.0);
INSERT INTO sales VALUES (18, 3, 11, '2025-01-10', 5, 21375.0);
INSERT INTO sales VALUES (19, 10, 10, '2025-01-11', 5, 26000.0);
INSERT INTO sales VALUES (20, 9, 9, '2025-01-12', 3, 21600.0);
INSERT INTO sales VALUES (21, 5, 13, '2025-01-12', 4, 15200.0);
INSERT INTO sales VALUES (22, 11, 12, '2025-01-16', 3, 72000.0);
INSERT INTO sales VALUES (23, 8, 13, '2025-01-18', 1, 3800.0);
INSERT INTO sales VALUES (24, 5, 6, '2025-01-20', 3, 37500.0);
INSERT INTO sales VALUES (25, 8, 5, '2025-01-21', 2, 60800.0);
INSERT INTO sales VALUES (26, 12, 3, '2025-01-21', 1, 72000.0);
INSERT INTO sales VALUES (27, 2, 13, '2025-01-23', 4, 14440.0);
INSERT INTO sales VALUES (28, 10, 11, '2025-01-23', 2, 9000.0);
```

```
INSERT INTO sales VALUES (29, 8, 11, '2025-01-23', 1, 4500.0);
INSERT INTO sales VALUES (30, 10, 6, '2025-01-23', 3, 33750.0);
INSERT INTO sales VALUES (31, 12, 10, '2025-01-24', 2, 9360.0);
INSERT INTO sales VALUES (32, 5, 7, '2025-01-24', 3, 79800.0);
INSERT INTO sales VALUES (33, 9, 9, '2025-01-27', 4, 28800.0);
INSERT INTO sales VALUES (34, 2, 4, '2025-01-28', 1, 61750.0);
INSERT INTO sales VALUES (35, 8, 6, '2025-01-30', 1, 12500.0);
INSERT INTO sales VALUES (36, 10, 2, '2025-01-30', 1, 98000.0);
INSERT INTO sales VALUES (37, 12, 10, '2025-01-31', 3, 15600.0);
INSERT INTO sales VALUES (38, 2, 1, '2025-01-31', 2, 270000.0);
INSERT INTO sales VALUES (39, 1, 5, '2025-02-01', 2, 60800.0);
INSERT INTO sales VALUES (40, 7, 6, '2025-02-02', 4, 50000.0);
INSERT INTO sales VALUES (41, 5, 13, '2025-02-03', 4, 14440.0);
INSERT INTO sales VALUES (42, 4, 12, '2025-02-03', 2, 43200.0);
INSERT INTO sales VALUES (43, 3, 12, '2025-02-04', 1, 24000.0);
INSERT INTO sales VALUES (44, 12, 13, '2025-02-05', 3, 11400.0);
INSERT INTO sales VALUES (45, 10, 7, '2025-02-06', 2, 53200.0);
INSERT INTO sales VALUES (46, 6, 13, '2025-02-07', 2, 7600.0);
INSERT INTO sales VALUES (47, 6, 9, '2025-02-07', 1, 7200.0);
INSERT INTO sales VALUES (48, 3, 5, '2025-02-10', 2, 60800.0);
INSERT INTO sales VALUES (49, 3, 7, '2025-02-11', 1, 26600.0);
INSERT INTO sales VALUES (50, 6, 5, '2025-02-12', 3, 96000.0);
INSERT INTO sales VALUES (51, 9, 12, '2025-02-12', 1, 22800.0);
INSERT INTO sales VALUES (52, 7, 8, '2025-02-13', 5, 82500.0);
INSERT INTO sales VALUES (53, 3, 9, '2025-02-13', 1, 7200.0);
INSERT INTO sales VALUES (54, 5, 6, '2025-02-13', 1, 12500.0);
INSERT INTO sales VALUES (55, 4, 13, '2025-02-15', 4, 15200.0);
INSERT INTO sales VALUES (56, 5, 6, '2025-02-15', 5, 62500.0);
INSERT INTO sales VALUES (57, 10, 4, '2025-02-16', 1, 65000.0);
INSERT INTO sales VALUES (58, 9, 10, '2025-02-17', 5, 26000.0);
INSERT INTO sales VALUES (59, 6, 6, '2025-02-17', 4, 50000.0);
INSERT INTO sales VALUES (60, 3, 7, '2025-02-17', 2, 56000.0);
INSERT INTO sales VALUES (61, 3, 10, '2025-02-17', 3, 14040.0);
INSERT INTO sales VALUES (62, 1, 9, '2025-02-19', 1, 7200.0);
INSERT INTO sales VALUES (63, 4, 11, '2025-02-21', 3, 12825.0);
INSERT INTO sales VALUES (64, 7, 8, '2025-02-21', 2, 31350.0);
INSERT INTO sales VALUES (65, 3, 11, '2025-02-21', 5, 22500.0);
INSERT INTO sales VALUES (66, 5, 5, '2025-02-23', 1, 30400.0);
INSERT INTO sales VALUES (67, 11, 11, '2025-02-23', 5, 22500.0);
INSERT INTO sales VALUES (68, 1, 7, '2025-02-24', 1, 28000.0);
INSERT INTO sales VALUES (69, 10, 11, '2025-02-24', 1, 4500.0);
INSERT INTO sales VALUES (70, 3, 12, '2025-02-25', 3, 68400.0);
INSERT INTO sales VALUES (71, 9, 6, '2025-02-25', 1, 11250.0);
INSERT INTO sales VALUES (72, 11, 8, '2025-02-26', 1, 16500.0);
INSERT INTO sales VALUES (73, 2, 9, '2025-02-27', 2, 13680.0);
INSERT INTO sales VALUES (74, 2, 6, '2025-02-27', 1, 12500.0);
INSERT INTO sales VALUES (75, 9, 6, '2025-03-01', 5, 59375.0);
INSERT INTO sales VALUES (76, 2, 12, '2025-03-02', 3, 68400.0);
INSERT INTO sales VALUES (77, 5, 4, '2025-03-03', 1, 65000.0);
INSERT INTO sales VALUES (78, 3, 12, '2025-03-03', 2, 43200.0);
INSERT INTO sales VALUES (79, 7, 11, '2025-03-04', 2, 9000.0);
INSERT INTO sales VALUES (80, 5, 3, '2025-03-05', 1, 68400.0);
INSERT INTO sales VALUES (81, 4, 1, '2025-03-05', 1, 135000.0);
INSERT INTO sales VALUES (82, 9, 1, '2025-03-06', 2, 270000.0);
INSERT INTO sales VALUES (83, 3, 6, '2025-03-07', 4, 50000.0);
INSERT INTO sales VALUES (84, 5, 4, '2025-03-07', 2, 130000.0);
INSERT INTO sales VALUES (85, 6, 9, '2025-03-08', 5, 36000.0);
INSERT INTO sales VALUES (86, 5, 9, '2025-03-08', 2, 12960.0);
INSERT INTO sales VALUES (87, 11, 12, '2025-03-08', 3, 68400.0);
INSERT INTO sales VALUES (88, 11, 6, '2025-03-09', 5, 59375.0);
INSERT INTO sales VALUES (89, 6, 10, '2025-03-10', 2, 10400.0);
INSERT INTO sales VALUES (90, 10, 11, '2025-03-10', 1, 4500.0);
INSERT INTO sales VALUES (91, 6, 13, '2025-03-13', 1, 3420.0);
INSERT INTO sales VALUES (92, 7, 7, '2025-03-13', 2, 50400.0);
INSERT INTO sales VALUES (93, 7, 12, '2025-03-15', 2, 48000.0);
INSERT INTO sales VALUES (94, 10, 9, '2025-03-16', 4, 27360.0);
INSERT INTO sales VALUES (95, 1, 9, '2025-03-16', 4, 28800.0);
INSERT INTO sales VALUES (96, 11, 3, '2025-03-16', 1, 72000.0);
INSERT INTO sales VALUES (97, 3, 1, '2025-03-18', 2, 243000.0);
INSERT INTO sales VALUES (98, 3, 6, '2025-03-18', 1, 12500.0);
INSERT INTO sales VALUES (99, 2, 8, '2025-03-21', 3, 47025.0);
INSERT INTO sales VALUES (100, 9, 3, '2025-03-21', 2, 136800.0);
INSERT INTO sales VALUES (101, 4, 12, '2025-03-23', 3, 68400.0);
INSERT INTO sales VALUES (102, 5, 13, '2025-03-23', 3, 11400.0);
INSERT INTO sales VALUES (103, 6, 4, '2025-03-24', 2, 117000.0);
INSERT INTO sales VALUES (104, 6, 6, '2025-03-26', 4, 45000.0);
INSERT INTO sales VALUES (105, 8, 4, '2025-03-26', 1, 65000.0);
INSERT INTO sales VALUES (106, 1, 11, '2025-03-26', 4, 18000.0);
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INSERT INTO sales VALUES (397, 12, 4, '2025-10-03', 2, 117000.0);
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INSERT INTO sales VALUES (400, 12, 8, '2025-10-03', 3, 47025.0);
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INSERT INTO sales VALUES (402, 6, 10, '2025-10-07', 3, 15600.0);
INSERT INTO sales VALUES (403, 10, 7, '2025-10-09', 2, 56000.0);
INSERT INTO sales VALUES (404, 5, 5, '2025-10-10', 1, 32000.0);
INSERT INTO sales VALUES (405, 3, 6, '2025-10-10', 2, 25000.0);
INSERT INTO sales VALUES (406, 6, 13, '2025-10-11', 3, 10830.0);
INSERT INTO sales VALUES (407, 7, 12, '2025-10-11', 2, 45600.0);
INSERT INTO sales VALUES (408, 12, 13, '2025-10-11', 2, 7600.0);
INSERT INTO sales VALUES (409, 9, 11, '2025-10-11', 3, 13500.0);
INSERT INTO sales VALUES (410, 1, 6, '2025-10-11', 4, 50000.0);
INSERT INTO sales VALUES (411, 9, 13, '2025-10-13', 1, 3800.0);
INSERT INTO sales VALUES (412, 12, 13, '2025-10-13', 4, 15200.0);
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INSERT INTO sales VALUES (415, 12, 10, '2025-10-14', 1, 5200.0);
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INSERT INTO sales VALUES (421, 6, 9, '2025-10-17', 4, 28800.0);
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INSERT INTO sales VALUES (428, 12, 13, '2025-10-27', 5, 19000.0);
INSERT INTO sales VALUES (429, 2, 10, '2025-10-30', 2, 10400.0);
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INSERT INTO sales VALUES (433, 5, 13, '2025-11-02', 3, 11400.0);
INSERT INTO sales VALUES (434, 6, 9, '2025-11-03', 2, 13680.0);
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INSERT INTO sales VALUES (437, 10, 8, '2025-11-05', 4, 62700.0);
INSERT INTO sales VALUES (438, 11, 10, '2025-11-06', 2, 9880.0);
INSERT INTO sales VALUES (439, 11, 8, '2025-11-06', 3, 49500.0);
INSERT INTO sales VALUES (440, 11, 13, '2025-11-07', 2, 7220.0);
INSERT INTO sales VALUES (441, 10, 10, '2025-11-08', 5, 24700.0);
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INSERT INTO sales VALUES (447, 1, 3, '2025-11-12', 1, 72000.0);
INSERT INTO sales VALUES (448, 5, 1, '2025-11-13', 1, 135000.0);
INSERT INTO sales VALUES (449, 8, 6, '2025-11-13', 3, 35625.0);
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INSERT INTO sales VALUES (451, 12, 5, '2025-11-14', 3, 96000.0);
INSERT INTO sales VALUES (452, 10, 3, '2025-11-15', 1, 72000.0);
INSERT INTO sales VALUES (453, 3, 7, '2025-11-16', 2, 56000.0);
INSERT INTO sales VALUES (454, 5, 11, '2025-11-16', 2, 9000.0);
INSERT INTO sales VALUES (455, 4, 13, '2025-11-17', 3, 11400.0);
INSERT INTO sales VALUES (456, 6, 10, '2025-11-17', 4, 20800.0);
INSERT INTO sales VALUES (457, 2, 13, '2025-11-18', 4, 15200.0);
INSERT INTO sales VALUES (458, 2, 3, '2025-11-18', 2, 144000.0);
INSERT INTO sales VALUES (459, 1, 7, '2025-11-19', 2, 56000.0);
INSERT INTO sales VALUES (460, 6, 9, '2025-11-19', 3, 21600.0);
INSERT INTO sales VALUES (461, 5, 13, '2025-11-19', 5, 18050.0);
INSERT INTO sales VALUES (462, 12, 5, '2025-11-20', 2, 57600.0);
INSERT INTO sales VALUES (463, 5, 10, '2025-11-20', 2, 10400.0);
INSERT INTO sales VALUES (464, 5, 11, '2025-11-21', 3, 13500.0);
INSERT INTO sales VALUES (465, 6, 13, '2025-11-22', 3, 11400.0);
INSERT INTO sales VALUES (466, 6, 11, '2025-11-25', 1, 4500.0);
INSERT INTO sales VALUES (467, 7, 9, '2025-11-27', 1, 6840.0);
INSERT INTO sales VALUES (468, 7, 13, '2025-11-27', 2, 6840.0);
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INSERT INTO sales VALUES (470, 2, 8, '2025-11-29', 2, 33000.0);
INSERT INTO sales VALUES (471, 1, 7, '2025-11-29', 2, 50400.0);
INSERT INTO sales VALUES (472, 5, 10, '2025-11-29', 4, 19760.0);
INSERT INTO sales VALUES (473, 10, 7, '2025-11-30', 3, 84000.0);
INSERT INTO sales VALUES (474, 1, 11, '2025-11-30', 5, 22500.0);
INSERT INTO sales VALUES (475, 4, 13, '2025-11-30', 4, 14440.0);
INSERT INTO sales VALUES (476, 5, 10, '2025-11-30', 2, 9360.0);
INSERT INTO sales VALUES (477, 3, 7, '2025-11-30', 3, 79800.0);
INSERT INTO sales VALUES (478, 6, 13, '2025-12-03', 2, 7600.0);
INSERT INTO sales VALUES (479, 8, 13, '2025-12-05', 4, 13680.0);
INSERT INTO sales VALUES (480, 9, 11, '2025-12-06', 2, 8550.0);
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INSERT INTO sales VALUES (489, 8, 8, '2025-12-12', 5, 78375.0);
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INSERT INTO sales VALUES (491, 1, 9, '2025-12-13', 5, 36000.0);
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INSERT INTO sales VALUES (500, 6, 8, '2025-12-18', 3, 44550.0);
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INSERT INTO sales VALUES (502, 12, 6, '2025-12-20', 5, 62500.0);
INSERT INTO sales VALUES (503, 1, 2, '2025-12-20', 2, 186200.0);
INSERT INTO sales VALUES (504, 4, 5, '2025-12-21', 1, 32000.0);
INSERT INTO sales VALUES (505, 9, 2, '2025-12-21', 1, 98000.0);
INSERT INTO sales VALUES (506, 10, 12, '2025-12-21', 1, 24000.0);
INSERT INTO sales VALUES (507, 6, 8, '2025-12-21', 1, 16500.0);
INSERT INTO sales VALUES (508, 4, 13, '2025-12-22', 5, 19000.0);
INSERT INTO sales VALUES (509, 12, 5, '2025-12-24', 3, 91200.0);
INSERT INTO sales VALUES (510, 7, 9, '2025-12-24', 4, 28800.0);
INSERT INTO sales VALUES (511, 7, 11, '2025-12-26', 3, 12825.0);
INSERT INTO sales VALUES (512, 8, 6, '2025-12-27', 4, 50000.0);
INSERT INTO sales VALUES (513, 2, 11, '2025-12-28', 5, 20250.0);
INSERT INTO sales VALUES (514, 11, 1, '2025-12-29', 2, 256500.0);
INSERT INTO sales VALUES (515, 9, 10, '2025-12-29', 3, 14820.0);
INSERT INTO sales VALUES (516, 3, 10, '2025-12-29', 5, 24700.0);
INSERT INTO sales VALUES (517, 8, 11, '2025-12-30', 5, 20250.0);
INSERT INTO sales VALUES (518, 8, 10, '2025-12-30', 2, 10400.0);
INSERT INTO sales VALUES (519, 5, 6, '2025-12-30', 3, 37500.0);
INSERT INTO sales VALUES (520, 11, 11, '2025-12-31', 5, 22500.0);

-- PART 3: ANALYSIS QUERIES

-- Q1. Sales rep performance against annual target
SELECT e.name, e.region,
       COUNT(s.sale_id)                     AS deals,
       ROUND(SUM(s.amount), 0)              AS total_sales,
       e.monthly_target * 12                AS annual_target,
       ROUND(100.0 * SUM(s.amount) / (e.monthly_target * 12), 1) AS achievement_pct,
       CASE WHEN SUM(s.amount) >= e.monthly_target * 12        THEN 'Exceeded'
            WHEN SUM(s.amount) >= e.monthly_target * 12 * 0.9  THEN 'Near target'
            ELSE 'Below target' END         AS status
FROM employees e
LEFT JOIN sales s ON s.emp_id = e.emp_id
GROUP BY e.emp_id
ORDER BY achievement_pct DESC;

-- Q2. Rank of each rep inside their region
SELECT region, name, total_sales,
       RANK() OVER (PARTITION BY region ORDER BY total_sales DESC) AS rank_in_region
FROM (
    SELECT e.region, e.name, ROUND(SUM(s.amount), 0) AS total_sales
    FROM employees e
    JOIN sales s ON s.emp_id = e.emp_id
    GROUP BY e.emp_id
)
ORDER BY region, rank_in_region;

-- Q3. Quarterly sales with cumulative total
WITH q AS (
    SELECT 'Q' || ((CAST(strftime('%m', sale_date) AS INTEGER) + 2) / 3) AS quarter,
           SUM(amount) AS sales
    FROM sales
    GROUP BY quarter
)
SELECT quarter,
```

```
       ROUND(sales, 0) AS sales,
       ROUND(SUM(sales) OVER (ORDER BY quarter), 0) AS cumulative
FROM q
ORDER BY quarter;

-- Q4. Best-selling product in each category
WITH product_sales AS (
    SELECT p.category, p.name AS product,
           SUM(s.quantity) AS units,
           ROUND(SUM(s.amount), 0) AS revenue
    FROM sales s
    JOIN products p ON p.product_id = s.product_id
    GROUP BY p.product_id
),
ranked AS (
    SELECT *, ROW_NUMBER() OVER (PARTITION BY category ORDER BY revenue DESC) AS rn
    FROM product_sales
)
SELECT category, product, units, revenue
FROM ranked
WHERE rn = 1
ORDER BY revenue DESC;

-- Q5. Stock alert: products at or below reorder level
SELECT p.name, p.category, p.stock, p.reorder_level,
       COALESCE(SUM(s.quantity), 0) AS units_sold_last_90_days
FROM products p
LEFT JOIN sales s
       ON s.product_id = p.product_id
      AND s.sale_date >= date((SELECT MAX(sale_date) FROM sales), '-90 days')
WHERE p.stock <= p.reorder_level
GROUP BY p.product_id
ORDER BY units_sold_last_90_days DESC;

-- Q6. Sales by day of the week
SELECT CASE strftime('%w', sale_date)
           WHEN '0' THEN 'Sunday' WHEN '1' THEN 'Monday' WHEN '2' THEN 'Tuesday' WHEN '3' THEN 'Wednesday' WHEN '4' THEN 'Thursday' WH
       END AS weekday,
       COUNT(*) AS deals,
       ROUND(SUM(amount), 0) AS sales
FROM sales
GROUP BY strftime('%w', sale_date)
ORDER BY strftime('%w', sale_date);

-- Q7. Products that were never sold
SELECT p.product_id, p.name, p.category, p.stock
FROM products p
WHERE NOT EXISTS (
    SELECT 1 FROM sales s WHERE s.product_id = p.product_id
);
```